

Engineering Chemistry Analysis

UNIT-WISE TOP 5 MOST IMPORTANT TOPICS

UNIT 1 — WATER TECHNOLOGY

1) Hardness of Water (Definition, Units, Types)

Years Asked: 2019 May, 2020 Dec, 2024 Dec

Marks: 7M–14M

► Always asked as: definition + units + relationship

2) Hardness Numericals (EDTA / CaCO_3 / Temporary–Permanent)

Years: 2019 May, 2020 Dec, 2022 Nov, 2023 Dec, 2024 Dec, 2025 June

Marks: 7M–14M

► MOST repeated numerical of Unit 1

3) EDTA Method (Principle + Procedure)

Years: 2020 Dec, 2022 Nov, 2023 Dec

Marks: 7M–10M

4) Lime-Soda Process (Softening + Numerical)

Years: 2022 June, 2023 June, 2023 Dec, 2025 June

Marks: 7M–14M

5) Ion-Exchange Process (With Diagram)

Years: 2020 June, 2022 June, 2022 Nov, 2023 Dec, 2025 June

Marks: 7M

Question Type	Frequency	How Asked
Definitions (Hardness, Units)	Very High	Always 1st part of long question
Numerical (Hardness / Alkalinity)	Extremely High	Most repeated numerical in whole syllabus
Softening methods (Lime-soda, Zeolite, Ion exchange)	High	Explain + Advantages
Short Notes	Medium	"Scales & sludges", "Caustic embrittlement"

UNIT 2 — BOILER TROUBLES & SOFTENING METHODS

1) Scales & Sludges (Formation + Differences)

Years: 2023 June, 2024 Dec, 2025 June

Marks: 7M

2) Caustic Embrittlement

Years: 2020 June, 2023 Dec, 2024 Dec, 2025 June

Marks: 5M–7M

3) Boiler Corrosion

Years: 2020 June, 2025 June

Marks: 7M–10M

4) Priming & Foaming

Years: 2020 June

Marks: 7M

5) Water-line & Galvanic Corrosion

Years: 2024 Dec, 2023 June

Marks: 5–7M

UNIT 2—Question Pattern

- Definition + causes + prevention
- Always part (a) or (b) of long questions
- Some years ask: **“Write short notes (Any 2)”** including:
 - Scales
 - Sludges
 - Caustic embrittlement
 - Boiler corrosion
 - Priming & foaming

UNIT 3 — LUBRICANTS & LUBRICATION

1) Viscosity Index

Years: 2020 Dec, 2022 June, 2023 June, 2023 Dec, 2024 Dec

Marks: 5M–7M

2) Flash Point & Fire Point

Years: 2020 Dec, 2022 June, 2023 Dec, 2025 June, 2023 June

Marks: 5M–7M

3) Mechanism of Lubrication (Thick / Thin Film)

Years: 2020 June, 2022 June, 2023 June, 2024 Dec

Marks: 7M

4) Aniline Point

Years: 2023 June, 2023 Dec, 2025 June

Marks: 5M–7M

5) Saponification Number / Emulsification Number

Years: 2023 June, 2022 June, 2024 Dec

Marks: 5M

UNIT 3—Question Pattern

- Define + explain significance
- Explain with neat diagram (for mechanism)
- Short notes frequently asked
- Numerical sometimes asked: viscosity index (2025)

UNIT 4 — POLYMERS & POLYMERIZATION

1) Types of Polymerization (Free Radical / Ionic / Step / Chain)

Years: 2020 Dec, 2022 Nov, 2023 June, 2024 Dec

Marks: 7M

2) PVC (Preparation, Properties, Uses)

Years: 2020 Dec, 2022 June, 2022 Nov, 2023 June

Marks: 7M

3) Bakelite / Phenol-formaldehyde Resin

Years: 2020 June, 2023 Dec, 2024 Dec

Marks: 7M

4) Nylon 6,6

Years: 2020 Dec, 2022 Nov, 2023 Dec

Marks: 5M–7M

5) Elastomers (Buna-S / Buna-N)

Years: 2022 June, 2023 June

Marks: 5–7M

UNIT 4—Question Pattern

- Preparation + properties + uses
- Mechanism questions very frequent
- Short notes on synthetic rubber
- Diagrams sometimes asked

UNIT 5 — PHASE RULE & CORROSION

1) Oxidation Corrosion (+ Mechanism)

Years: 2020 Dec, 2022 June, 2023 June, 2024 Dec, 2025 June

Marks: 7M–14M

2) Phase Rule (Terms + Diagrams)

Years: 2020 Dec, 2023 June, 2025 June

Marks: 7M–14M

3) Chemical vs Electrochemical Corrosion

Years: 2023 June

Marks: 7M

4) Water-line / Pitting / Stress Corrosion

Years: 2023 June, 2024 Dec

Marks: 5M–7M

5) Triple Point (Phase Diagram)

Years: 2022 June, 2023 Dec

Marks: 5M–7M

UNIT 5—Question Pattern

- Draw diagrams
- Explain significance
- Full-length mechanism questions
- Short notes on corrosion types

UNIT 6 — SPECTROSCOPY

1) Beer–Lambert Law

Years: 2019 Nov, 2020 Dec, 2023 Dec

Marks: 5M–7M

2) Principle, Instrumentation & Applications (IR / UV / Electronic)

Years: 2020 June, 2020 Dec, 2022 Nov, 2023 June, 2024 Dec, 2025 June

Marks: 7M–14M

3) Electronic Transitions / Shifts (Bathochromic, Hypsochromic)

Years: 2023 Dec

Marks: 5M

4) Vibrational & Rotational Spectroscopy

Years: 2022 Nov, 2023 June

Marks: 7M

5) Structure Determination Using IR + UV

Years: 2025 June

Marks: 7M

UNIT 6—Question Pattern

- Almost always a full 7M or 14M question
- Theory-heavy

- Sometimes asks comparison (UV vs IR)
- Short notes appear in “Any Two” section

UNIT 7 — PERIODIC PROPERTIES

1) Atomic & Ionic Size

Years: 2020 Dec, 2023 Dec, 2024 Dec, 2025 June

Marks: 5M–7M

2) Electronegativity

Years: 2023 Dec, 2024 Dec, 2025 June

Marks: 5M

3) Electron Affinity

Years: 2023 June, 2023 Dec, 2025 June

Marks: 5M

4) Effective Nuclear Charge

Years: 2025 June

Marks: 5M

5) Oxidation States

Years: 2025 June

Marks: 5M

UNIT 7—Question Pattern

- Explanation + Trend + Example
- Sometimes diagrams
- Often asked as short notes (Any Two / Any Three)

UNIT-WISE EXPECTED QUESTIONS FOR THIS YEAR

UNIT 1 — WATER TECHNOLOGY (HIGH WEIGHTAGE)

EXPECTED QUESTIONS

1 EDTA method – Principle + Hardness Calculation (Numerical)

Most repeated combination in all years.

2 Lime-Soda Process (Explain + Numerical)

Repeated: 2022, 2023, 2025

→ Expected again.

3 Ion Exchange Method (Diagram + Working)

If Lime-soda numerical appears, this theory will appear.

4 Alkalinity Numerical (Phenolphthalein + Methyl Orange)

Asked many times; returns every 2–3 years.

5 Temporary vs Permanent Hardness (Definition + Examples)

Appears whenever EDTA is not asked.

UNIT 2 — BOILER TROUBLES & SOFTENING

EXPECTED QUESTIONS

1 Scales & Sludges – Formation + Prevention

Most stable repeating question.

2 Caustic Embrittlement

Asked in 2020, 2023, 2024, 2025 → highly likely.

3 Boiler Corrosion – Mechanism + Prevention

When caustic embrittlement is asked as short note, this becomes long question.

4 Priming & Foaming

Appears every alternate exam.

5 Zeolite Process (if asked in Unit 1, here comes as short note)

UNIT 3 — LUBRICANTS & LUBRICATION

EXPECTED QUESTIONS

1 Viscosity Index (Definition + Significance)

Top 3 repeated in this unit.

2 Flash Point & Fire Point (Difference + Significance)

Very frequent.

3 Mechanism of Lubrication (Thin Film / Thick Film)

Asked in 2020, 2022, 2023, 2024 → highly expected.

4 Aniline Point

Repeats frequently.

5 Saponification Number / Emulsification Number

Usually a short note.

UNIT 4 — POLYMERS & POLYMERIZATION

EXPECTED QUESTIONS

1 Free Radical vs Ionic Polymerization (Mechanism)

Always repeated.

2 PVC – Preparation, Properties & Uses

Highest repeated polymer.

3 Bakelite (Phenol-formaldehyde Resin)

Asked many times → Expected again.

4 Nylon 6,6 (Preparation + Uses)

Repeats in almost every 2 years.

5 Elastomers (Buna-S / Buna-N)

If PVC isn't asked, this one comes.

UNIT 5 — PHASE RULE & CORROSION

EXPECTED QUESTIONS

1 Oxidation Corrosion (Mechanism + Prevention)

Biggest repeated long question of this unit.

2 Phase Rule – Definition + Terms + Diagram of Water System

Always asked in 2–3 year cycles.

3 Chemical vs Electrochemical Corrosion

Mostly short notes.

4 Water-line, Pitting, Stress Corrosion

Frequent short note.

5 Triple Point

Appears every 2–3 years, highly likely.

UNIT 6 — SPECTROSCOPY

EXPECTED QUESTIONS

1 Beer–Lambert Law (Statement + Derivation)

Very frequently asked.

2 IR Spectroscopy – Principle, Instrumentation, Applications

Most repeated spectroscopy question.

3 UV / Electronic Spectroscopy

(Principle + Transitions) expected as long question.

4 Vibrational & Rotational Spectroscopy

Often appears when IR is not asked.

5 Chromophores + Shifts (Bathochromic / Hypsochromic)

Asked in 2023, expected again.

UNIT 7 — PERIODIC PROPERTIES

EXPECTED QUESTIONS

1 Atomic & Ionic Size – Trends + Explanation

Top repeated.

2 Electronegativity (Group & Period Trend)

Asked recently → likely again.

3 Electron Affinity

Frequently asked.

4 Effective Nuclear Charge (Z_{eff})

Trend + definition expected.

5 Oxidation States

Very common short note.

10 MOST REPEATED QUESTIONS OF ALL TIME (From 2019–2025)

RANK 1 — EDTA Method + Hardness Numerical

Every 1–2 years, extremely repeated.

RANK 2 — Lime-Soda Process + Numerical

Top numerical question of the subject.

RANK 3 — Viscosity Index

Appears in almost every exam.

RANK 4 — Flash Point & Fire Point

Very stable repeated short/long answer.

RANK 5 — Mechanism of Lubrication (Thick / Thin Film)

Major repeated theory.

RANK 6 — PVC / Nylon 6,6 / Bakelite

One polymer question is ALWAYS present.

RANK 7 — Oxidation Corrosion (Mechanism)

Strongest repeated long question in corrosion.

RANK 8 — Phase Rule + Water System Diagram

Frequently asked every two years.

RANK 9 — Beer–Lambert Law

Guaranteed appearance in some form.

 RANK 10 — Periodic Trends (Size/Electronegativity/Electron Affinity)

At least ONE trend question in every exam.

FULL EXAM PAPER PREDICTION – MODEL PAPER 2025

based on 2019–2025 PYQ

ENGINEERING CHEMISTRY — MODEL PAPER (HIGHLY PREDICTIVE)

BT-101 (GS) – FULL EXAM MODEL PAPER (LIKELY 2025 PATTERN)

Time: 3 Hours | **Max Marks:** 70

Note: Attempt *any five* questions. All questions carry equal marks.

Q1. WATER TECHNOLOGY (Unit 1)

a) Explain the EDTA method for determination of hardness of water.

Discuss the **principle**, **reaction**, **indicator**, and **end point**.

b) A 100 mL water sample required 18 mL of N/50 EDTA solution.

Calculate the **total hardness** in terms of CaCO_3 equivalent.

Q2. BOILER TROUBLES (Unit 2)

a) What are scales and sludges? Describe their formation, harmful effects, and prevention methods.

b) Explain caustic embrittlement. What steps are taken to minimize this boiler trouble?

Q3. LUBRICANTS & LUBRICATION (Unit 3)

a) What is viscosity index? How is it determined?

Explain its **importance** for evaluating lubricant quality.

b) Define flash point and fire point.

Explain their **significance** in selecting suitable lubricants.

Q4. POLYMERS (Unit 4)

a) Explain the mechanism of free radical polymerization with neat steps.

Compare it with **ionic polymerization**.

b) Write preparation, properties, and uses of PVC or Bakelite.

Q5. CORROSION & PHASE RULE (Unit 5)

a) What is oxidation corrosion?

Explain the **mechanism**, **factors affecting it**, and **preventive measures**.

b) State the Phase Rule equation.

Draw and explain the **phase diagram of water system**, marking its curves and triple point.

Q6. SPECTROSCOPY (Unit 6)

a) State and explain the Beer–Lambert Law.

Discuss its **limitations** and **applications** in chemical analysis.

b) Explain the principle, instrumentation, and applications of IR spectroscopy.

Q7. PERIODIC PROPERTIES (Unit 7)

a) Explain the periodic trends of:

- i) **Atomic size**
- ii) **Ionic size**
- iii) **Electronegativity**

b) What is effective nuclear charge?

Explain its role in periodic variations of elements.

Q8. SHORT NOTES SECTION (ANY TWO)

Write short notes on any **two** of the following:

- i) **Alkalinity of water**
- ii) **Aniline point**
- iii) **Buna-S / Buna-N Rubber**
- iv) **Pitting corrosion**